

Lunbing Chen

Bio-inspired Flight · Dynamic Soaring · Vortex Dynamics · Autonomous Control

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Research Profile

I am a Ph.D. candidate in [J. C. Wu Center for Aerodynamics](#) at [Shanghai Jiao Tong University](#). My research focuses on **closed-loop dynamic soaring and bio-inspired flight in complex environmental flows**. I combine unsteady aerodynamics, animal flight, reinforcement learning, and autonomous aircraft systems to understand how flyers exploit environmental flows for robust and energy-efficient flight.

My current thesis develops dynamic soaring as a feedback-driven energy-harvesting navigation problem, connecting passive flow mechanisms, biological flight strategies, step-level feedback decision-making, and closed-loop flight implementation.

Research Interests

- **Dynamic soaring and energy-harvesting flight:** optimal trade-offs, flight envelopes, and target-directed navigation in shear flows [1, 2, 3].
- **Learning-based autonomous flight:** reinforcement-learning policies and feedback structures for step-level dynamic soaring [2].
- **Bio-inspired robust flight:** passive aerodynamic stability, disturbance attenuation, and avian wing mechanisms [4, 5].
- **Unsteady vortex dynamics:** vortex growth, transient force generation, and low-order modeling of accelerating bodies [6, 7, 8, 9, 10, 11].

Education

Shanghai Jiao Tong University

Ph.D. Candidate, J. C. Wu Center for Aerodynamics

2022–Present

Shanghai, China

- Dissertation topic: *Closed-loop Dynamic Soaring in the Field: Mechanics and Implementation*
- Advisors: Prof. [Yang Xiang](#) and Prof. [Hong Liu](#)

Shanghai Jiao Tong University

B.Eng., School of Aeronautics and Astronautics

2018–2022

Shanghai, China

Research Experience

Closed-loop Dynamic Soaring and Energy-harvesting Navigation

2022–Present

Dynamic soaring, optimal control, reinforcement learning, autonomous flight

Investigate dynamic soaring as a closed-loop energy-harvesting navigation problem in environmental shear flows. The work connects optimal-control models, biological flight trajectories, and reinforcement-learning policies for robust target-directed flight.

- Formulated dynamic soaring as a trade-off between energy harvesting and directional progress.
- Built 3-DOF gliding-flight models and numerical optimization frameworks for shear-flow flight.
- Developed deep reinforcement-learning environments for step-level dynamic-soaring navigation.
- Identified feedback structures that coordinate turning, vertical motion, and energy management.

Related publications: [1, 2, 3]

Passive Flow Mechanisms for Robust Gliding

2022–Present

Flow–structure interaction, passive reconfiguration, turbulence, aerodynamic sensitivity

Study how passive structural and aerodynamic mechanisms improve flight robustness in unsteady and disturbed flows. This work connects canonical experiments with biological wing aerodynamics to understand how flyers reduce load peaks and attenuate disturbance amplification before active control is required.

- Investigated accelerating reconfigurable circular plates as a canonical model for passive load regulation.
- Identified load-shifting mechanisms in which flexible structures reduce peak loads and reshape post-peak force histories.
- Compared avian wings with engineered airfoil wings in turbulent inflow to quantify aerodynamic sensitivity and passive stability.

Related publications: [6, 4, 5]

Unsteady Vortex Dynamics and Force Generation

2021–2025

Vortex rings, impulse theory, accelerating bodies, PIV, low-order force modeling

Investigated vortex-ring growth, impulse-based force generation, and low-order modeling of starting and accelerating flows.

- Conducted force measurements on accelerating plates and related unsteady bluff-body flows.
- Used time-resolved PIV to characterize vortex-ring formation, vortex growth, and wake evolution.
- Connected transient force histories with circulation growth, vortex area evolution, and spatial vorticity distribution.

Related publications: [6, 9, 10, 11]

Closed-loop Dynamic Soaring Aircraft Development

2024–Present

UAV platform, state estimation, flight control, system integration

Developing a closed-loop dynamic-soaring aircraft platform that integrates sensing, decision-making, and execution.

- Designed a perception–decision–execution architecture for wind-relative energy management.
- Integrated onboard sensing for position, attitude, airspeed, altitude, and wind-relative state estimation.
- Implemented decision and control modules for step-level maneuver selection, attitude tracking, safety protection, and mode switching.

Related publications: [2]

Earlier Aerospace Engineering Projects

2019–2022

Uncertainty quantification, numerical simulation, engineering software

Worked on undergraduate aerospace engineering projects involving numerical modeling, thermochemical ablation, and software development.

- Studied high-dimensional uncertainty quantification of C/C nozzle throat ablation.
- Developed a mobile-based spacecraft management application for status monitoring and task publishing.

Related publications: [12]

Publications

First-author / Corresponding-author Publications

- [1] Lunbing Chen, Yufei Yin, Yang Xiang, Suyang Qin, and Hong Liu. “Optimal dynamic soaring trades off energy harvest and directional flight”. In: *iScience* 28.6 (2025). DOI: [10.1016/j.isci.2025.112540](https://doi.org/10.1016/j.isci.2025.112540).
- [2] Lunbing Chen, Jixin Lu, Yufei Yin, Jinpeng Huang, Yong Chen, Yang Xiang, and Hong Liu. *Learning step-level dynamic soaring in shear flow*. Under external review at *Nature Communications*. 2026. DOI: [10.48550/arXiv.2604.12413](https://doi.org/10.48550/arXiv.2604.12413). arXiv: [2604.12413](https://arxiv.org/abs/2604.12413).

- [4] Lunbing Chen, Suyang Qin, Qilin Wu, Jinpeng Huang, Yufei Yin, Yang Xiang, and Hong Liu. “Reduced aerodynamic sensitivity enables disturbance-resilient flight”. Submitted to *Cell Reports Physical Science*. 2025. DOI: [10.48550/arXiv.2507.14550](https://doi.org/10.48550/arXiv.2507.14550).
- [6] Lunbing Chen, Zhuoqi Li, Qilin Wu, Yufei Yin, Yang Xiang, and Hong Liu. “On the load-shifting phenomenon of accelerating reconfigurable circular plates”. In: *Journal of Fluid Mechanics* 1029 (2026), A8. DOI: [10.1017/jfm.2026.11165](https://doi.org/10.1017/jfm.2026.11165).
- [7] Jinpeng Huang, Qilin Wu, Lunbing Chen*, Yang Xiang, and Hong Liu. “Static Geometric Configuration and Vortex Ring Reshaping Dominate the Load-shifting of Accelerating Plates”. Submitted to *Physics of Fluids*. Lunbing Chen is the corresponding author. 2026.
- [8] Qilin Wu, Lunbing Chen*, Yang Xiang, and Hong Liu. “Shape Independence and Aerodynamic Truncation Effect of Low-Aspect-Ratio Accelerating Plates”. In preparation for *Physical Review Fluids*. Lunbing Chen is the corresponding author. 2026.
- [12] Lunbing Chen, Na Sun, Linying Li, and Bin Zhang. “High-Dimensional Uncertainty Quantification of Thermochemical Ablation for C/C Throat Insert”. In: *Journal of Propulsion Technology* 43.3 (2022). (in Chinese), pp. 266–275. DOI: [10.13675/j.cnki.tjjs.200583](https://doi.org/10.13675/j.cnki.tjjs.200583).

Selected Co-authored Publications

- [3] Yufei Yin, Lunbing Chen, Yang Xiang, Suyang Qin, and Hong Liu. “Flight Envelope and Energy Equilibrium Strategy of Sailplanes to Perform Dynamic Soaring”. In: *International Conference on Aerospace System Science and Engineering*. Springer, 2024, pp. 17–27. DOI: [10.1007/978-981-96-2440-9_3](https://doi.org/10.1007/978-981-96-2440-9_3).
- [5] Jinpeng Huang, Yang Xiang, Lunbing Chen, Suyang Qin, Jixin Lu, Sen Ye, Xiaobin Huang, Yong Chen, and Hong Liu. “Geese achieve stationary takeoff via synergistic wing kinematics and enhanced aerodynamics”. Under review in *Science Advances*. 2025. DOI: [10.48550/arXiv.2512.20894](https://doi.org/10.48550/arXiv.2512.20894).
- [9] Zhuoqi Li, Lunbing Chen, Yang Xiang, Hong Liu, and Fuxin Wang. “Two independent mechanisms with distinct laws for the generation of drag force on accelerating plates”. In: *Physics of Fluids* 36.10 (2024). DOI: [10.1063/5.0227258](https://doi.org/10.1063/5.0227258).
- [10] Yufei Yin, Lunbing Chen, Liwei Hu, Yang Xiang, Yumin Zhong, and Hong Liu. “Evolution and dynamics of vortex rings in a confined and asymmetric region”. In: *Physics of Fluids* 37.12 (2025). Editor’s pick, Scilight. DOI: [10.1063/5.0304637](https://doi.org/10.1063/5.0304637).
- [11] Bin Yu, Lunbing Chen, Zhen Wang, Xunchen Liu, Bin Zhang, and Hong Liu. “Study on the structural characteristics of strong radial swirl flow based on Lagrangian and Eulerian vortex identification methods”. In: *CCTAM 2021*. (in Chinese). 2021.

Technical Skills

Programming: Python, MATLAB, Fortran, Java, LaTeX, Git

Numerical Modeling: ODE integration, direct collocation, trajectory optimization, nonlinear dynamics, low-order modeling

Machine Learning and Control: Deep reinforcement learning, Gym-style simulation environments, feedback policy analysis, autonomous control

Experimental Fluid Mechanics: Force measurement, towing-tank experiments, wind-tunnel experiments, PIV/SPIV, high-speed imaging

Aerodynamics and Flight Mechanics: 3/6-DOF glider dynamics, stability analysis, unsteady aerodynamics

Data Analysis: Signal processing, uncertainty analysis, trajectory analysis, image processing, statistical visualization